nature machine intelligence

Perspective

https://doi.org/10.1038/s42256-025-01146-z

What neuroscience can tell AI about learning in continuously changing environments

Received: 1 February 2025

Accepted: 7 October 2025

Published online: 28 November 2025

Check for updates

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Modern artificial intelligence (AI) models, such as large language models, are usually trained once on a huge corpus of data, potentially fine-tuned for a specific task and then deployed with fixed parameters. Their training is costly, slow and gradual, requiring billions of repetitions. In stark contrast, animals continuously adapt to the ever-changing contingencies in their environments. This is particularly important for social species, where behavioural policies and reward outcomes may frequently change in interaction with peers. The underlying computational processes are often marked by rapid shifts in an animal's behaviour and rather sudden transitions in neuronal population activity. Such computational capacities are of growing importance for AI systems operating in the real world, like those guiding robots or autonomous vehicles, or for agentic AI interacting with humans online. Can AI learn from neuroscience? This Perspective explores this question, integrating the literature on continual and in-context learning in AI with the neuroscience of learning on behavioural tasks with shifting rules, reward probabilities or outcomes. We outline an agenda for how the links between neuroscience and AI could be tightened, thus supporting the transfer of ideas and findings between both areas and contributing to the evolving field of NeuroAI.

Natural environments are never constant. Food sources may be depleted and novel, even more nutritious opportunities may pop up. Strategies for escaping from harmful situations that worked in one context may fail in another. Changing weather conditions, seasons or migration into novel terrain may impose new challenges[1,2]. This is even more true in interaction with predators, potential prey and conspecifics, who may change their own behavioural policies in response to yours, leading to multi-level feedback loops[2]. Animal brains evolved over hundreds of millions of years to deal with such challenges[3]. This ability to continuously adapt to changing environmental conditions and to come up with new solutions on the fly, especially in dangerous and time-critical situations, is a characteristic of increasing importance to AI[4]. It is of obvious relevance to robots and autonomous agents directly interacting, like animals, with the physical world. It, however, also has an increasing role for AI systems engaging with humans online, like medical support systems or other large language model (LLM)-based assistants and 'agentic AI'[5].

The process of training current state-of-the-art machine learning (ML) and AI models fundamentally differs, however, from how animals learn. Unlike animals, they are usually trained once on a huge corpus of data[6]. After this initial training, there is comparatively little flexibility in readjusting parameters to cope with novel scenarios. The training itself is slow and gradual, commonly based on forms of gradient descent (GD). This is strikingly different from animal brains, which continuously absorb new knowledge, and may ingrain one-time-only experiences forever within the physical structure of their synaptic connectivity[7,8].

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Although recent LLMs such as DeepSeek V3.1 or OpenAI's o3 showcase impressive reasoning skills (but see ref. 9), the versatility and flexibility with which small animal brains adapt to non-stationary environments is still unmet by today's most advanced AI systems.

Here we explore how mechanisms proposed in (computational) neuroscience on how animal brains deal with non-stationary environments may inspire AI research. We start with a brief overview of the ways in which current AI models adjust to novel tasks and statistical shifts. Unlike previous reviews on the topic[10–13], we not only discuss strategies for continual and lifelong learning in ML but also integrate this literature with the more recent phenomenon of in-context learning. In-context learning provides another means by which AI models may quickly respond to novel challenges. In line with this, in our review of related neuro-behavioural research, we put a focus specifically on non-stationary tasks that are similar in design to those used for probing continual and in-context learning in AI. Animal brains may master such tasks through two basic mechanisms—neuro-dynamical and synaptic. Synaptic mechanisms and their relation to continual learning have been discussed before[10,11], so we particularly highlight more recent findings such as behavioural timescale plasticity (BTSP)[14]. Neuro-dynamical mechanisms and their potential role in continual and in-context learning, however, have received little attention so far. Both these mechanisms in animal brains crucially differ from current AI algorithms and reflect adaptation to the multiple demands and timescales of their non-stationary environments. We conclude with a discussion of how neuroscience findings could be transferred more directly into AI development, and how—vice versa—progress in ML and AI may fertilize and guide neuroscience.

Learning in non-stationary environments in AI systems

Fundamentally, there are two ways of dealing with novel situations that profoundly differ from what an agent has experienced before (referred to as out-of-domain or out-of-distribution (OOD)). It can adapt its parameters to incorporate these new experiences, sometimes dubbed 'in-weights' learning[15,16]. In ML and AI, this includes any form of parameter fine-tuning and algorithms for continual and lifelong learning. Or it could try to infer the optimal response from observations online, related to what has been called 'in-context' learning[17–20], which includes methods of prompting to induce reasoning processes, such as chain-of-thoughts[21]. We start by reviewing these two areas in ML and AI.

Adapting to changing environments through continual learning

AI systems are usually trained through iterative numerical schemes, most commonly some form of GD. Common optimizers such as stochastic GD (SGD) or RAdam[22] refine the basic GD process by injecting stochasticity or using adaptive learning rates. GD, like essentially any numerical optimization scheme, is a slow gradual procedure, progressing with learning rates on the order of 10^{-3}–10^{-6} (for example, ref. 23). It requires—depending on model and data size—up to billions of repetitions and training samples for the process to converge[18,24]. While GD techniques have been tremendously successful and represent the de facto standard for training ML and AI models, they are thus computationally costly and data intense[6]. Retraining a full model as more data become available is computationally prohibitive. Moreover, there is the problem of catastrophic forgetting[25–27]: as a model is retrained on new tasks, it tends to forget previous solutions, which is fatal if old tasks reappear in the model's environment (Fig. 1a). The problem of maintaining adaptivity while preserving previous skills has also been dubbed the 'plasticity–stability dilemma'[28,29]. It has been a known issue for decades, almost since the serious study of neural networks took off in the 1980s[28,30,31]. Numerous ways of addressing it have been suggested, including early loosely bioinspired solutions such as adaptive resonance theory[28]. Modern ML and AI approaches essentially fall into four major categories[32,33] (excellent previous reviews of this field, including connections to biology, are available[10,11,34], so we will be brief here). First are solutions that prohibit strong shifts in network parameters to prevent catastrophic forgetting[26,35,36]; second, architectural solutions such as modularity or freezing part of the structure in later learning[37–40]; third, replay of previous content interspersed with new tasks[41,42]; and, fourth, functional solutions such as partial network resets to maintain plasticity[43]. To test the capability for improved continual learning, certain benchmarks were defined in the field. Fundamentally, these consist of an initial training on a set of tasks, and then continual training on new tasks with task identity provided (task-incremental learning), new domains or contexts (domain-incremental learning), or new classes without identifying the switches between different tasks as such (class-incremental learning)[33,44].

Regarding the first approach, the idea is to penalize strong shifts in parameter space that would erase previously learned tasks. This could be implemented through regularization terms that keep updated parameter estimates close to previously successful solutions and/or within a Bayesian setting that places a strong prior on previous parameters[26,35,36]. Regarding the second approach, a standard strategy for fine-tuning LLMs on new data is keeping part of the system frozen and allowing changes to only some of the parameters. This can happen in various ways, for example, making only the final layers of a network trainable or inserting new modules ('adapters')[45], or through 'low-rank adaptation', which decomposes weight matrices into fixed (previously trained) and low-rank trainable parts[46]. Another strategy that particularly helps transfer to new domains is modularity, with different network modules learning different subtasks[39,47]. There are also architectures that endorse explicit memory (storage) mechanisms, such as neural Turing machines[48] or retrieval-augmented transformers[49], which thereby could alleviate catastrophic forgetting. These have issues with scalability and stability, however[50,51]. Experience replay, the third class of approaches, is a procedure inspired by the phenomenon of replay first observed in the rodent hippocampus, where during sleep (but also awake states, as we know now) previously visited sequences are re-instantiated in neural activity[52,53]. Experience replay schemes mix new task learning with selective presentation of old class samples in an optimal way[41,42]. An example of the fourth strategy is the recent work by Dohare et al.[43] on continual backpropagation (Fig. 1a). Here, a small percentage of network units with only little contribution to the activity of other units is continuously reset, that is, re-initialized with new weights, keeping the network alive. This is based on the insight that during continual learning 'unit debris' starts to pile up, that is, many units saturate and no longer contribute, thus reducing the flexibility of the system to adapt to new tasks.

While all these methods reduce or—more or less—abolish the catastrophic forgetting problem, they are mostly still based on GD training and hence inherit its natural limitations: learning is comparatively slow, gradual and costly, requiring many repetitions and a large number of samples. Moreover, the focus of most of these methods is on eradicating catastrophic loss of previous memories. Only a few approaches, such as refresh learning[32], have shifted attention towards embracing new information. Either way, updating is usually neither rapid nor satisfied by just a few samples, in stark contrast to biological learning[8,54]. Rather, in modern AI, anything based on few samples (few- or zero-shot learning) or quick updating is left to online inference, discussed next.

Adapting to changing environments through inference

LLMs and LLM-based reasoning models show an apparently remarkable capability for OOD generalization. The term OOD most commonly refers to shifts in the statistical distribution of the test set[55] (in contrast to the classical statistical out-of-sample generalization[56]), but may also refer to other strong shifts in tasks or conditions, such as changes in the dynamical regime a time series comes from[57]. A particular form

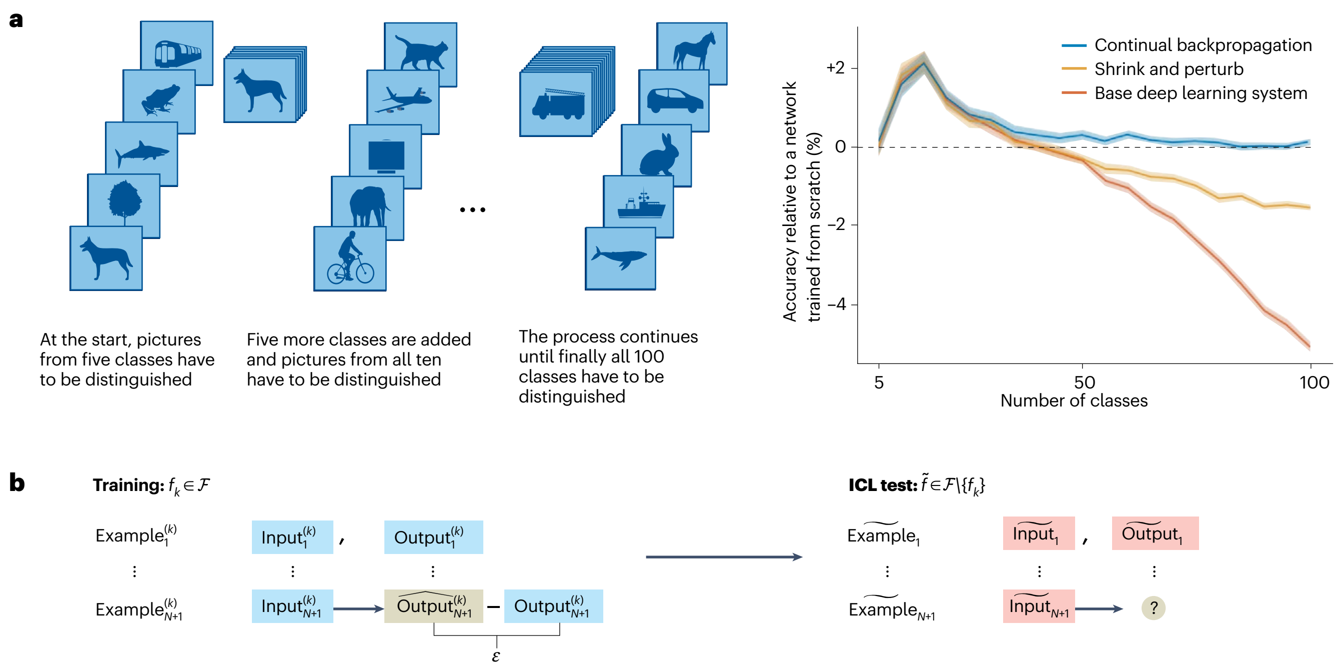

Fig. 1 | Continual and in-context learning. a, Left: in class-incremental CIFAR, new image classification tasks are continuously added to the previous repertoire of already learned classifications. Right: for common deep learning systems, performance steadily decays below that of networks trained from scratch as new classification tasks keep on being introduced. Continual backpropagation is one recently introduced training algorithm to amend this problem. **b**, In-context learning refers to the observation that large foundation models can generalize to new functions without any model retraining, just by being presented with examples from a new function 'in-context'. During training (left), the network (mostly transformers) is presented with a sequence of examples of different input–output pairs from a function f_k randomly drawn from some general function class F, for example, linear or nonlinear regression problems in Garg et al.[20]. The network is tasked to predict the last output corresponding to the last presented input, and the loss for training (indicated by ε) is evaluated on these last pairs from each sampled function (left). During inference (right), a new function $\tilde{f}$ not included in the training set $\{f_k\}$ is drawn from the same general function class (for example, a set of new regression weights in a linear regression problem), from which a sequence of examples is provided in-context. No weight learning happens during this phase. The network then makes a prediction on the last presented input, surprisingly often performing almost as well as a model explicitly trained from scratch on these new pairs of input–output values[20]. Panel **a** reproduced from ref. 43 under a Creative Commons licence CC BY 4.0.

of few-shot OOD generalization in LLMs has been dubbed 'in-context learning' (ICL)[18,20]. The term 'learning' might be slightly confusing here, as there is actually no learning in terms of parameter retraining or fine-tuning. Instead, the model is presented with a sequence of samples $\{x_i, y_i\}$, $i = 1, \ldots, N$, from a new task or function not contained in the model's training set. It then infers the correct answer on a probe item x_{N+1} (Fig. 1b). Zero-shot refers to the case '$N = 0$', that is, if the model is able to infer the correct output for a new task directly just from the instruction or type of query, as in reasoning processes elicited by 'chain-of-thought' prompts[21,58].

As this phenomenon is hard to analyse in massively pre-trained LLMs, partly because properties of the training corpus are not always clear, recent studies tried to isolate it by training transformers or other foundation architectures from scratch[15,20,59,60]. For instance, in Garg et al.[20], transformers were trained on a set of regression problems $\{x_{i,k}, y_{i,k} = f_k(x_{i,k})\}$, $i = 1, \ldots, N$, $k = 1, \ldots, K$, and then during test time were shown a sequence of samples $\{\tilde{x}_i, \tilde{y}_i = \tilde{f}(x_i)\}$, $i = 1, \ldots, N$, from a new function $\tilde{f}$ not contained in the training set, that is $\tilde{f} \notin \{f_k\}$ (Fig. 1b). The mechanisms underlying ICL are still not well understood. A strong claim here is that ICL rests on a kind of 'in-context GD', based on proofs by construction that successive transformer layers could implement GD steps[17,61]. However, this idea relies on strong assumptions about the model's weight matrices, which are empirically not met, nor do the output distributions produced by GD and ICL agree[62–64]. Also, it is hard to conceive how millions of GD steps needed for more challenging tasks could be implemented this way. A perhaps more mundane, but at the same time more biology-friendly, explanation could be that ICL simply relies on retrieving close examples through a process resembling recall in associative memory[9]. Other hypotheses are that ICL interpolates among training samples[65], utilizes compositional structure[66] or recombines tasks learned in pre-training[63].

With in-context GD an unlikely explanation for the observed ICL capabilities, ICL remains limited by the properties of the training distribution[9,65,67]. Training sets therefore need to be massive, and new data or tasks, violating the training distribution, seem hard to accommodate. Moreover, even test-time compute can be expensive and comparatively slow[68]. The brain has found other solutions, as we illuminate in the following, where new learning and inference are often tightly interwoven.

Learning in non-stationary environments in animals

The literature on animal learning is extensive and spans a broad spectrum, from the behaviourist principles of classical and operant conditioning[69] to ideas in modern cognitive ethology and psychology about mental simulations and learning world models[70]. Rather than re-reviewing this whole field, our focus here is specifically on learning in non-stationary tasks, involving shifting options, rules or reward values, which are homologous, in a certain sense, to common AI benchmarks. From these, we distil core observations that we deem most relevant to AI.

Many learning tasks in neuroscience are close in design to those used for benchmarking AI systems, in the sense that consecutive samples of 'input–output pairs $\{x_i, y_i\}$, $i = 1, \ldots, N$' (trials) are presented. The inputs are commonly (but not necessarily) discrete sets of stimuli, and the outputs are requested behavioural choices. However, the correct output is rarely explicitly conveyed to the animal, but only indicated through a reward signal, making most tasks in neuroscience

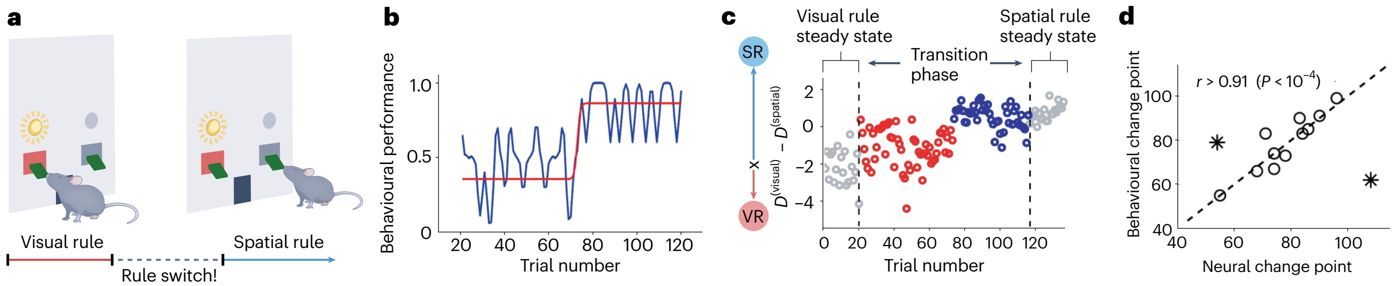

Fig. 2 | Sudden transitions in behaviour and neural population representations during rule learning. a, Animals are first trained on a 'visual' operant conditioning rule, and then—unknowingly—switched to a different rule in which the previous cue needs to be ignored and instead place (lever site) becomes relevant for reward (that is, task contingencies change and the animal has to infer this by trial and error). **b**, At some point after the experimental rule switch, performance of individual animals rather abruptly jumps from chance level to nearly perfect performance (gradual learning curves as often seen in the literature are an 'artefact' of averaging across many such jumps[86]). **c**, These transitions in behaviour are accompanied by rather abrupt transitions in the neural population activity. The difference between the distances $D^{(\text{visual})}$ and $D^{(\text{spatial})}$ of the neural population vectors to the rule-specific steady states (VR, visual rule; SR, spatial rule) is shown as a function of trial number. **d**, The trials at which these rapid transitions are observed can be statistically inferred through change point analysis, and change points observed in neural activity and behavioural performance tightly correlate (except for two outliers, indicated by stars, with the weakest evidence for change points). Figure reproduced with permission from ref. 88, Elsevier.

reinforcement learning paradigms. An (operant) rule defines the correct mapping from inputs to outputs. In some tasks, this mapping is probabilistic, mimicking the uncertainty in the world. In rule learning or shifting tasks (Fig. 2a), more specifically, animals are first trained to perform on one rule, and after achieving a certain performance level, task rules (reward contingencies) are changed unbeknown to the animal. Rule-shifting tasks are an example of what would be considered a continual learning paradigm in ML and AI, with novel rules introduced consecutively, but old rules repeated once in a while[71,72]. They are more in line with class- rather than task-incremental learning, as task identity is not provided to the animal—it must be inferred by sampling from the environment. There are also paradigms with explicit context cues indicating the active rule, more similar to task-incremental learning[73]. Context is important for behavioural flexibility, allowing an animal to adjust its behaviour depending on other environmental variables[34,73,74].

If one focuses on a particular rule switch, one may also see rule shifting as an ICL paradigm: given a sequence of observed trial outcomes, the animal needs to infer the new task rule. In ML and AI, continual learning depends on parameter changes, while ICL is an inference process not requiring in-weights learning. In animals, there is no clear distinction, and behaviour on rule-shifting tasks may involve either one or a mixture of both. The exact trade-off may depend on an animal's previous exposure to various rules[71] and natural biases acquired throughout evolutionary history, such as 'win–stay, lose–shift'[75], similar to ICL's dependence on a model's training corpus[9,18]. Thus, the neural mechanisms may differ depending on whether rules are learned de novo, or whether they are already in the animal's repertoire and just revisited, a property that may not be immediately evident to the animal from the environmental input[71,76]. Although it may not always be possible to clearly distinguish what an animal has encountered before and what is really novel, laboratory animals are usually raised in highly controlled environmental settings and the tasks they are exposed to are often quite remote from their natural habitats.

Other task paradigms that involve continuous changes in the animal's environment are drifting n-armed bandit and gambling tasks[77–79]. In these tasks animals or humans are confronted with a set of choice options. After choosing an option, a reward is delivered. Different choice options vary in the size or probability of reward, and the task is to learn which option leads to the largest reward. Learning is engaged by periodically changing the choice options in various ways. For instance, the reward probability associated with each option can vary over time, as exemplified by experiments with rodents in Passecker et al.[78]. Here, rats could make a choice between two options in a Y-arm maze, one yielding a low probability high reward and the other a high probability low reward. After a block of trials, the reward probability of the 'gamble' option was altered, without this change being explicitly signalled to the animals. All these task manipulations require individuals to update their choice policies continuously. One difference to rule learning and shifting tasks is that the rules of the game do not change, and this restricts the solution space. Rule-shifting and drifting bandit tasks are considered markers of 'cognitive flexibility'[80,81], the ability to swiftly adapt to changing environmental conditions, and heavily rely on brain structures such as the prefrontal cortex and striatum[81] and their neuromodulation by dopamine[82].

A couple of observations about the neuronal and behavioural processes in such tasks are potentially relevant to AI. First, even if a rule is new, it rarely takes an animal thousands of repetitions to learn it, but often a couple of trials are sufficient. Second, animals continue to explore other options even when the task rules are stable[69]. This is a reasonable prior in non-stationary environments where reward contingencies may change once in a while. Third, performance relies on previously learned task elements, rules, natural biases, task schemata and concepts[40,83–85]. This compositional reuse of previously learned schemata and behavioural building blocks is thought to be essential for quick adaptation and generalization[83–85]. In fact, procedures for animal training in the lab usually consist of several stages of familiarizing the animal with the different task elements ('shaping')[69], reminiscent of curriculum learning protocols in ML and AI. Fourth, behaviour does not change gradually throughout the task, but often there are sudden performance jumps[86,87] (Fig. 2b). These jumps in behavioural performance are accompanied by rapid transitions in neuronal representations in prefrontal cortex and other brain areas[71,88,89] (Fig. 2c). Such jumps are observed even if a rule is novel and has not previously (explicitly) been introduced to an animal[88]. Rapid performance and neuronal changes also mark the onset of exploratory phases, when an animal realizes that a previous behavioural strategy no longer works[88,90]. Fifth, in tasks where a previous response is extinguished ('extinction learning'), that is, is no longer rewarded, the previous behaviour is not unlearned nor forgotten, but suppressed[69]. For instance, extinguished behaviours can be rapidly reinstated once the reward is reintroduced[69]. Interestingly, extinction learning, like acquisition of a new rule, goes hand in hand with sudden transitions in behaviour and neuronal population activity[91].

Taken together, these observations indicate that animal learning is an active and compositional process: animals systematically explore different behavioural strategies, which adaptively recombine previously learned task elements, rather than incrementally acquiring stimulus-response contingencies as posited by classical behaviourist

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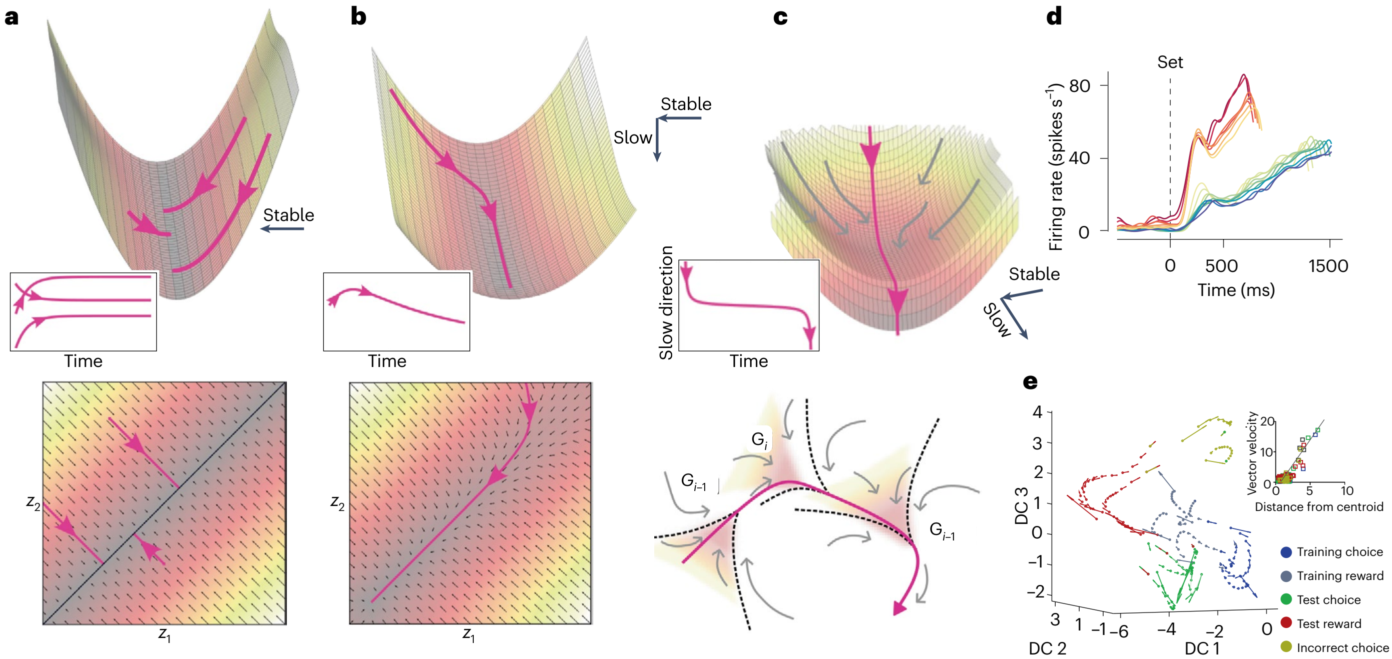

Fig. 3 | Dynamical mechanisms and their experimental support. a, Manifold attractors, here illustrated in a two-dimensional state space (bottom graph) spanned by two dynamical variables z_1 and z_2, are continuous sets of marginally stable fixed points, towards which the flow is directed from all sides as indicated by the system's vector field in grey (which gives the direction and magnitude of change at each point in state space as determined by the system's differential equations). The flow velocity is further indicated by the background shading, corresponding to the steepness of the valleys in the pseudo-potential above (top graph). Formally, fixed (or equilibrium) points are points in state space at which the flow of all state variables vanishes, that is, the velocity vector becomes zero, as on the black line (bottom of the valley). Trajectories (pink) from three different initial conditions converge to a point on the line (manifold) attractor, which thereby stores a memory of the initial condition. The pseudo-potential above the state space provides another illustration of the line attractor configuration, with trajectories traveling downhill until they come to rest anywhere at the flat floor of the valley. **b**, As the manifold attractor is slightly detuned (bottom graph), flow velocity continuously changes from zero, creating a 'ghost line', a region of arbitrarily slow flow along which trajectories slowly crawl to a stable point attractor. In the pseudo-potential (top graph) this corresponds to a slight tilt of the floor of the valley, such that system states now slowly roll along it. This phenomenon can be used to create arbitrary timescales in the system along the 'ghost manifold'. Insets: the time graphs associated with these trajectories. **c**, A ghost attractor (pseudo-potential in the top graph) is an attractor that just lost stability, such that in its vicinity the flow is arbitrarily slow (as indicated by the time graph in the inset, and similar to **b**). Trajectories enter the ghost attractor region (as visualized by the grey and pink arrows) from almost all directions but, different from **b**, slowly leave along one direction again. Bottom: different such ghost attractors G_i may be arranged in a ghost attractor chain, a construction that can flexibly implement sequences. **d**, Ramping firing rates flexibly adjusting to the length of a temporal interval between different events (in this case a set signal and a motor response), as observed in various brain areas[138,139], are consistent with the idea of such ghost manifolds. Reddish ('short') and blueish ('long') colours indicate two different interval lengths; the different curves within each of these two classes correspond to firing rates associated with slight variations in the produced interval. **e**, Experimental support for ghost attractor constructs in the nervous system comes from neural recordings during multiple-item working memory and decision-making tasks. A neural population activity state space reconstructed from multiple single-unit recordings with estimated flow vectors along trajectories is shown, illustrating slowing down (quantified in the inset) as the system's neural population state enters different task stages (colour-coded). DC, discriminant coordinate. Panels adapted with permission from: **a**,**b**, ref. 129, ICLR; **c**, ref. 136, APS. Panels reproduced from: **d**, ref. 139 under a Creative Commons licence CC BY 4.0; **e**, ref. 137 under a Creative Commons licence, CC BY 4.0.

theories[71,92]. They actively explore and probe their environments to reduce uncertainty and update internal models[93]. While active inference[94] and directed information seeking are strategies that have been adopted to some degree in the field of reinforcement learning in ML and AI[95–100], there is still a stark contrast to how most neural network training is done in ML, which is largely passive and incremental. Two classes of mechanisms may support rapid and flexible adaptation in animal brains, dynamical and plasticity mechanisms, which we discuss in turn.

Neuro-dynamical mechanisms supporting adaptation to non-stationary environments

Dynamical systems theory (DST) provides a general mathematical framework for understanding the behaviour of natural or engineered systems evolving in time, formalized as systems of differential equations or recursive maps. DST has been a mainstay in computational and theoretical neuroscience for decades, serving well for explaining computation in the nervous system and how it evolves from the underlying physiology. Dynamical systems are Turing complete[101–103]. While DST is increasingly recognized also in ML and AI as a toolbox for explaining the behaviour of transformers[104,105], recurrent neural networks (RNNs)[106,107] or training algorithms such as SGD[108–111], how dynamics could be utilized to perform computation has been less appreciated in the AI field. Notable exceptions are Hopfield networks[112] and Boltzmann machines[113,114], whose attractor dynamics support robust memory storage, retrieval and pattern completion. Diffusion models exhibit similar dynamical mechanisms[115,116]. The perspective DST provides on computation in the brain could be of major relevance for agentic AI and online learning in natural environments, partly because it puts the emphasis on the temporal aspects and many timescales on which biological and physical processes unfold.

DST deals with the topological and geometrical properties of state spaces (Fig. 3), subsets of Euclidean space that contain all states a dynamical system could be in (that is, the set of values the dynamical variables can attain). The temporal evolution of a system corresponds to a trajectory through state space (Fig. 3), eventually converging to geometrical objects called attractors, which could be single points (fixed points or equilibria), limit cycles (corresponding to stable oscillations) or objects with more complicated, fractal geometry, called chaotic attractors. Computational neuroscience conceives computational processes as being implemented in terms of such trajectories, with their fate ruled by the state space's geometrical properties. There is substantial experimental support for the existence of various types of attractor objects in the brain and their role in computation[117–119].

One important concept is that of a manifold attractor[120–124], a continuous set of marginally stable equilibria towards which the system's state evolves from various initial conditions, while on the manifold itself there is no further movement unless the system is pushed by external inputs or noise (Fig. 3a). A manifold attractor provides a computational template for 'long short-term' (working) memory: external inputs guide the system's trajectory to a unique state on the manifold, storing that information indefinitely if unperturbed (Fig. 3a). This is in the absence of any parameter changes. Empirical evidence for manifold attractors in the brain abounds[117–119,125]. One classical example is the ability of fish and mammals to maintain arbitrary eye positions[117,124,126]. Physiologically this seems to be supported by a nearly continuous set of stable neural firing rates, evident even at the single-cell level[127]. Another example is the nearly continuous representation of place or head direction in the rodent hippocampus[118,120]. Causal interventions by electrophysiological[117,118] or optogenetic[128] techniques have specifically probed predictions of the manifold attractor idea, making this one of the best supported theoretical concepts in neuroscience[118,119].

A specific implementation in AI systems can be found in Schmidt et al.[129], where manifold attractors were encouraged in RNN training through regularization terms in the loss function, gently pushing network weights towards configurations that support directions of zero flow in latent space. RNNs equipped with these dynamical mechanisms were shown to outperform most other architectures, such as long short-term memory networks[130], on typical long-range arena problems. At the same time, this allowed to keep the RNN design simple and thus more interpretable and easier to train. Manifold attractors can also integrate continuous variables such as reward feedback in a bandit task, thus adapting the system to changing environments without requiring synaptic plasticity[79]. In some sense, manifold attractors are a dynamical substrate for a long context window, yet potentially much less resource intensive than context windows in transformer-based models.

The brain may engage many much more complex dynamical constructs to realize computational properties, and even computational neuroscience has hardly scratched the surface here. Separatrix cycles[131], or heteroclinic channels[132,133], are sets of saddle nodes (or cycles) connected through so-called heteroclinic orbits that lead from one node to the next. These allow for robust yet flexible implementation of sequences, in which elements can quickly be exchanged. Hence, they also enable fast adaptation to novel contexts, which, for instance, require novel structuring of a series of motor commands[132]. Relatedly, chains of ghost attractors have been proposed to serve a similar purpose[134–136]. These are remnants of attractors that have lost their stability, so are not formally attractors anymore, but still pull in trajectories from a larger surrounding that then slow down as they move into the ghost's immediate vicinity (Fig. 3c). Neural activity observed in the rat anterior cingulate cortex during multiple-item working memory and decision-making tasks has properties consistent with such ghost attractor chains, with trajectories considerably slowing down as animals reach decision points in the task[137] (Fig. 3e). Attractor ghosts can also explain the observation of ramping neuronal activity profiles with adjustable slopes in timing tasks[123,138,139] (Fig. 3b,d). The idea of chaotic itinerancy, the chaotic wandering among ghost attractors, takes this one step further and has been suggested to underlie flexible and adaptive cognitive operations[135].

In RNN-based AI systems, attractor ghosts have been exploited to map processes evolving on arbitrarily slow timescales, by learning to adapt a ghost manifold to the timescales observed in the data[129] (Fig. 3b). Ghost attractor chains have also been suggested to provide a natural mechanism for replaying relevant memories and preventing catastrophic forgetting in attractor neural networks[140]. As attractor ghosts hover just at the threshold towards stability, in conjunction with fast plasticity rules or neuromodulation they also provide a means to quickly rearrange temporal sequences in order with changing task demands[141,142]. Slight and transient changes in weight space or neuromodulatory tone[82,143] are sufficient to drive certain states above the threshold while drowning others, thus quickly and flexibly altering the flow among motor primitives, concepts or memory items. More generally, incorporating biological temporal dynamics and resonance properties has been observed to yield advantages for target localization in drones[144] to spatio-temporal pattern processing[145]. A recent brain-inspired reasoning model[146], by mimicking dynamics on different timescales, achieves state-of-the-art performance on various complex reasoning tasks at only a fraction of the parameter and memory load of LLMs. Similar low-cost performance boosts have recently been observed in foundation models for time series built on dynamical systems principles[23].

Another important concept in DST is that of a bifurcation, related to phase transitions in physics or tipping points in many complex natural, for example, climate, systems. A bifurcation is a qualitative, topological change in a system's state space as one or more of its parameters are changed[131]. Many types of bifurcations incur abrupt changes in the system dynamics as one of the attractor objects suddenly vanishes and/or novel objects appear. Bifurcations may drive the restructuring of dynamical motifs supporting different behavioural rules[84], accounting for the sudden jumps observed in neuronal representations and behaviour in the rule-shifting tasks discussed above[88]. Interestingly, such sudden jumps also occur during GD-based training of recursive or autoregressive AI models, where they empirically and formally could be linked to bifurcations[108,147]. Systems harbouring ghost attractors are close to multiple bifurcations in parameter space, and, more generally, there is some evidence that brains operate in near-bifurcation regimes[148–151]. This may explain their versatility in rapidly adjusting to novel situations: even fairly small changes in parameters could lead to a fundamental reorganization of the system's computational structure for a novel purpose at hand. Physiologically, such fast restructuring via bifurcations may be regulated through neuromodulators such as dopamine, which has been implied in controlling prefrontal cortical attractor landscapes in relation to working memory and cognitive flexibility[82,143].

A crucial take-home here is that dynamical mechanisms not only allow for flexible and fast reconfiguration in novel surroundings but also involve a natural notion of time and timescales absent in modern AI systems: the world's non-stationary evolution is mirrored by the brain's non-stationary dynamics[152].

Plasticity mechanisms supporting adaptation to non-stationary environments

In the brain, changes in synaptic function, but also cellular changes in the contribution of different voltage-gated ion channels to neuronal activity[153], are thought to underlie learning and behavioural adaptation[7]. In the following, we focus on two specific properties of synaptic plasticity that fundamentally distinguish it from in-weights learning in AI. First, plasticity is organized on many different levels and across multiple timescales. Second, the brain's plasticity mechanisms are largely unsupervised[154], continual, and support one-shot and incidental learning.

Regarding the first point, there are multiple mechanisms and forms of synaptic plasticity, operating on many different timescales, with differential expression across brain areas[7,155]. There is synaptic short-term depression, facilitation, post-tetanic potentiation and augmentation, within hundreds of milliseconds to possibly minutes. Forms of long-term depression and long-term potentiation may set in within minutes, and last up to hours, days or longer[7,155]. Synaptic plasticity may involve structural changes, such as changes in the morphology

 https://doi.org/10.1038/s42256-025-01146-z

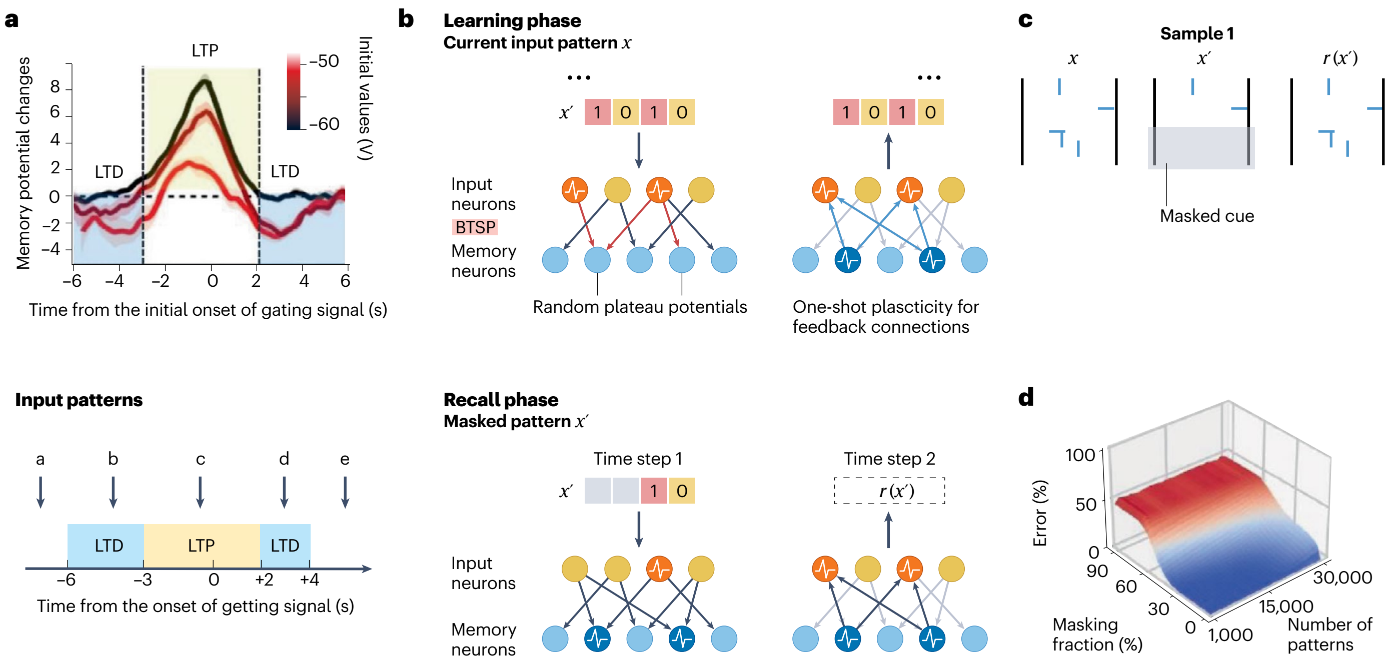

Fig. 4 | Fast plasticity mechanisms supporting episodic memory. a, BTSP (top) and its simplified formalization (bottom). BTSP depends on both the temporal lag between synaptic inputs and a dendritic plateau potential (x axis), and the initial strength of the synaptic weight (colour coding), with a weight increase induced if synaptic input and dendritic depolarization coincide within a time window of a few seconds, and a weight decrease otherwise. Physiologically, a weight increase is measured as a long-lasting increase in the amplitude of post-synaptic potentials, called 'long-term potentiation' (LTP), whereas a decrease is associated with 'long-term depression' (LTD). **b**, Top: by implementing such a BTSP mechanisms in a two-layer network, in which dendritic plateau potentials are induced in memory neurons at random times, different patterns can be instantaneously stored. Retrieval (completion of an original input pattern at the input layer) is accomplished by one-shot Hebbian (correlational) learning of feedback connections (right). Bottom: through these feedback connections, the originally observed pattern can be fully restored on the input layer from partial (occluded or noisy) cues, leading to memory retrieval similar as in a Hopfield network. **c**, Example of recovery, $r(x')$, of the original input pattern x from a partially occluded cue x'. **d**, Error rate for downstream classification of recalled patterns as a function of memory load (number of stored patterns) and the fraction of input bits masked. Figure reproduced from ref. 8 under a Creative Commons licence CC BY 4.0.

of synaptic boutons, dendritic spines and possibly arbors, the formation of new or the removal of existing synapses[156]. Remarkably, even such structural changes may set in within minutes to hours during or after a learning experience[157,158]. Similar to weight normalization in ML[159,160], global synaptic scaling rules ensure that neuronal activity does not run wild, as in epilepsy[155,161]. These many forms and types of synaptic plasticity once again highlight that in biological systems the temporal dimension is crucial—physical, biological and societal systems dynamically evolve in time, across many different scales, a feature reflected in physiological priors. Moreover, plasticity is itself subject to modulation by other factors, so-called meta-plasticity[162,163]. For instance, neuromodulators such as dopamine also regulate synaptic plasticity in line with motivational states and environmental factors such as prediction errors and novelty[164–166]. On the slowest timescale, the potential for plasticity itself undergoes massive changes throughout a lifetime, following specific developmental trajectories. While all cortical areas retain a capacity for plasticity throughout life, in primary sensory and motor systems the developmental window closes early, with cognitive systems following later[167,168]. This may allow the brain to efficiently build on stable sensory and motor representations when learning more complex tasks[40,71,168].

Some of these properties already feature in rudimentary form in AI systems. Adding fast weights to RNNs can boost performance on long-range associative memory tasks[169]. A large diversity of learning rates improves accuracy and robustness in deep neural networks and speeds up learning[170]. Similarly, equipping (synaptic) weights[162,171] or eligibility traces[172] with a broad range of biological time constants helps continual learning[162,171] or solving temporal credit assignment problems across a variety of delays[172]. Brain-inspired meta-learning strategies[173] to control which weights are subject to updating[163,174,175], how to best initialize them[4] or the type of plasticity mechanism itself[176,177], can alleviate catastrophic forgetting, foster rapid and continual adaptation to new tasks, and support the formation of schemata[178]. Developmental plasticity windows, finally, are mirrored by strategies and curriculum learning protocols that differentially regulate the plasticity and learning rates of different network layers[179–182].

Second, synaptic plasticity is ongoing and continual. There are no separate training and testing or deployment phases. In fact, a particularly remarkable ability of animals and humans is to store away one-time-only experiences on the fly and recall details of it days, weeks or even years later, even in the absence of explicit positive or negative reinforcers. This has been termed latent or incidental learning[183,184]. While unsupervised, latent learning and episodic memory, potentially in conjunction with experience replay, may also support (semi-supervised) reinforcement learning by allowing solution of the credit assignment problem across potentially long temporal gaps[154,185]: details that seem innocuous now may become relevant later.

But how does the brain manage to continuously and quickly absorb new information without disrupting existing memory contents? Part of the story likely is that different brain areas are differentially engaged in various forms and stages of learning and memory, which led to the hypothesis of complementary learning systems as a means to prevent catastrophic forgetting[34,186–188]. Complementary memory systems might be set up to minimize the risk of overwriting precious memory contents while serving fast updating, and for balancing a need for detail with the formation of abstract, generalizable concepts[34,187,188]. In particular the hippocampus plays a crucial role here, the brain structure best characterized from the cellular and synaptic up to the behavioural levels in terms of plasticity, learning and memory[7]. One basic idea is that the hippocampus functions as a fast memory buffer that over the

Table 1 | Translation of neuroscientific phenomena into AI

Neuroscience phenomenon	Presumed biological function	AI implementation (examples)
Manifold attractors	Resource-efficient, integrated, graded working memory not requiring parameter updating	• Encouraging manifold attractors in RNN latent space through regularization terms in loss function[129]
Ghost attractors, ghost channels, heteroclinic channels	Adaptive realization of arbitrary timescales; rapid, flexible rearrangement of temporal sequences; relevance-dependent replay of memory contents	• Loss regularization adapting manifold ghosts in RNN latent space to map multiple timescales in data[129] • Flexibly adjustable sequences of motor primitives, or switching between motor programmes, in robot control[141,142] • Allows for transient 'pop-out' of relevant memory representations in attractor neural networks, preventing catastrophic forgetting[140]
Synaptic plasticity on multiple timescales	Capturing natural phenomena evolving on multiple timescales; allowing for rapid updates and adaptation without overwriting memory; balances flexibility and stability	• Multiple timescale synaptic plasticity provides efficient memory scaling and fast updating without catastrophic forgetting[162,171] • Multiple timescale eligibility traces enable solving temporal credit assignment problems across variety of delays[172] • Distribution of weight-specific learning rates improves accuracy and robustness while speeding up learning in deep neural networks[170]
Structural plasticity	Reallocates capacity for new tasks while preserving core functions; supports long-term learning, robustness and flexibility	• Weight rejuvenation via activity-dependent re-initialization and weight splitting promotes continual adaptation and speeds up learning while improving robustness in various deep networks[43,170]
Developmental plasticity stages	Flexible adaptation by compositional combination of lower-level sensory and motor concepts into higher-level schemata; reducing interference	• Dynamical selection of network layers to freeze based on their stability and importance during training improves continual learning[181]
Metaplasticity	Regulates synaptic plasticity itself in accordance with environmental demands, current state of the organism and biological relevance	• Selecting weights subject to updating, weight re-initializations or adapting parameters of the learning rule itself fosters fast adaptation to novel tasks while preventing catastrophic forgetting[4,163,174,176,177]
Complementary learning systems	Allows for rapid incidental and episodic learning without interfering with or disrupting existing memory structures; enhances formation of generalizable schemata	• Optimizes the trade-off between generalization (extracting general rules) versus rapid encoding of episodic details by combining feed-forward networks with auto-associative memory networks in a student–teacher configuration[187,188], or combining artificial and spiking neural networks[195]
BTSP	Rapid one-shot encoding of behaviourally relevant events, serving episodic memory and incidental learning	• Defining two plasticity time windows with unsupervised weight-dependent stochastic update rule supports rapid, one-shot formation of high-capacity, content-addressable associative memories in artificial neural networks, robust to overwriting[8]

course of a day collects experiences and integrates them into episodic memory[189–191]. Daytime experiences are then replayed to the neocortex during sleep[192,193] for careful integration into existing memory structures and schemata, preventing their catastrophic forgetting[85,187]. While experience replay is well established in continual learning[41,42,194], the complementary memory systems hypothesis for mitigating catastrophic forgetting received renewed interest recently, with interesting directions combining artificial and spiking neural networks.[195]

In line with the unsupervised nature of incidental learning, a lot of plasticity in the nervous system is unsupervised and local, although semi-supervised or supervised forms of plasticity, involving forms of backpropagation, exist as well[154,196,197]. This may allow for considerable speed-ups and fewer resources in training, as no global error signals need to be propagated through the whole system. Historically, loosely bioinspired unsupervised (Hebbian) learning rules like those featured in self-organizing maps[198,199], or Oja's rule[200,201], actually had a big role in neural network research[202–204]. While these have gone out of fashion with the advance of modern GD techniques, revisiting them combined with GD may thus have a big potential. Many forms of longer-term plasticity further seem to be set up to capture causal or predictive relationships within the constant stream of daily episodic input: synaptic plasticity often depends on the precise spiking (action potential) patterns among pre- and post-synaptic neurons, so-called spike-timing-dependent plasticity[7]. This asymmetry in spike-time dependency may help to ingrain predictive relationships between environmental events[166,205,206]. Causality is obviously important for biological beings interacting with the physical world. Moreover, learning rules inspired by spike-timing-dependent plasticity can dramatically cut the computational costs associated with GD-based techniques[207–209], and lend themselves directly to implementation in energy-efficient neuromorphic hardware[210].

Another key aspect about incidental learning is that it is rapid, with a possible cellular substrate for it identified recently—BTSP[7,14,54,211,212]. BTSP, first examined in hippocampal place cells, sets in within seconds and requires only one or a few expositions to create a lasting memory of an environmental event[14,54,211]. Translated into artificial neural networks, BTSP supports the rapid, one-shot formation of high-capacity, content-addressable associative memories, which are also more robust to overwriting than other mechanisms[8,213] (Fig. 4). It is thus conceivable that such rapid forms of synaptic plasticity also provide mechanisms for in-context learning, where task representations are updated from just a few examples.

Conclusion and future directions

Natural environments, from physical to societal systems, are highly dynamic, often non-stationary, involving distribution shifts and tipping points into different dynamical regimes. They evolve across timescales from milliseconds to years, exhibiting anything from simple oscillatory to complex, multi-level, hyper-chaotic dynamics created by dense nets of interwoven feedback loops. Unlike current AI systems, brains are set up to parse, abstract and exploit this rich temporal structure for predicting future outcomes and preparing for changing conditions, processes reflected in physiological priors and the brain's large intrinsic dynamical repertoire resonating with the real world[152,214–221]. Brains continuously absorb, store and process new information, do so on the fly, incidentally, unsupervised, without explicit reward feedback, and without overwriting but rather suppressing currently irrelevant contents. They actively seek and filter relevant information, test hypotheses and constantly restructure memory[7,71,155,222]. These processes are supported by an array of synaptic and dynamical mechanisms that operate on a multitude of timescales and tightly interweave learning and inference. In contrast, current

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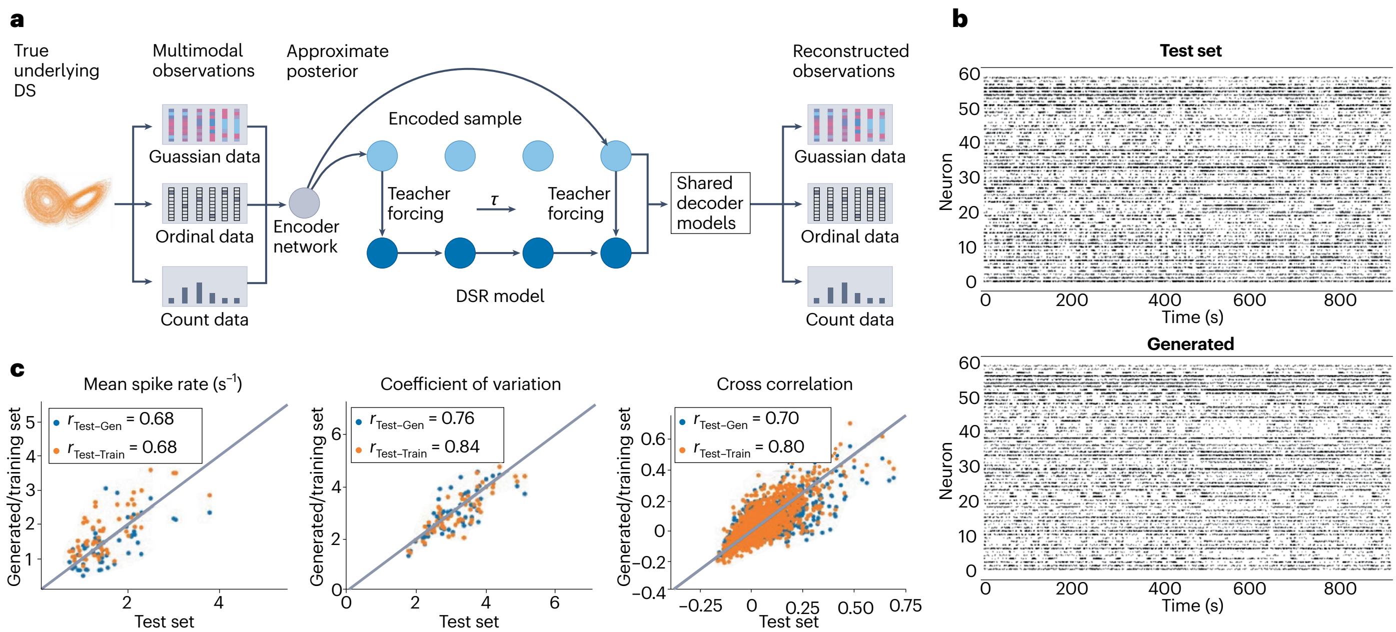

Fig. 5 | Learning computational surrogate models from neuronal and behavioural data. a, Multimodal teacher forcing architecture. In multimodal teacher forcing, a DSR model, like an RNN, is controlled in training by a multimodal teacher signal (applied every τ time steps) derived from an encoder network (like a temporal convolutional neural network) that integrates across different measurement modalities with different statistical properties (like spike count and ordinal behavioural response data) produced by the same underlying dynamical system (DS). The DSR model and the multimodal variational autoencoder are coupled to the same set of modality-specific decoder models (with same parameters), which map the latent states onto the observations, ensuring consistency between their latent codes. **b**, Examples of true (top) and DSR model-simulated (bottom) spike trains on an independent test set, from which only the initial condition was inferred to seed the DSR model. **c**, Agreement in different spiking statistics between true and model-generated spiking activity (blue) is about as good as between different segments of the real data (orange), indicating that the model became a faithful surrogate of the real system within the bounds of statistical fluctuations. Figure reproduced with permission from ref. 235, PMLR.

AI systems largely lack this flexibility and inherent representation of multi-scale temporal dynamics[9].

In the previous sections, we gave a number of examples of how some of the discussed biological properties were translated into ML and AI systems, improving performance and/or reducing computational costs. The most prominent of these are summarized in Table 1. Mostly, these aimed to mimic the brain's solution strategies, computational mechanisms or learning principles at a more abstract level. A literal translation of biological parameters and process details into AI systems is not feasible or sensible, not least because the brain and AI architectures differ so widely and biophysical details become meaningful only within the whole brain context., There may, however, still be ways of integrating neuroscience and AI research more tightly. This starts with a much tighter alignment of behavioural and cognitive task designs in both areas, which could help to transport neurophysiological findings more directly into AI design, or—vice versa—utilize knowledge about how AI systems solve tasks to explain actual data. For example, ML and AI algorithms for continual learning have so far been benchmarked mostly with incremental classification tasks, like the task-incremental Canadian Institute For Advanced Research (CIFAR) dataset (for example, refs. 43,44). It is not clear whether these are actually the most useful testbeds for the real-world scenarios for which continual learning procedures are designed. Task designs used in animal research may offer more realistic and ecologically appropriate use cases, especially relevant for agentic AI. Vice versa, benchmarks in ML and AI are created to probe specific functionalities needed to advance methods. Hence, how animals would solve similar tasks could give insight into neural mechanisms that underlie precisely these computational functions. Likewise, in both animals and AI models, properties of an artificial training corpus could be systematically varied to check for OOD generalization and its dependence on previous learning history.

AI could also lend a systematic, functional-computational perspective that could guide research questions in neuroscience in a specific way. For instance, research about principles of synaptic plasticity may incorporate ideas about potential optimization objectives that would yield specific testable hypotheses. A popular strategy in computational neuroscience recently is to train RNNs on the same types of task utilized in animal experiments, analyse how and by which dynamical principles they solve the task, and use these insights to interpret the neurophysiological and behavioural observations[74,83,223–225]. While these are most commonly simple, vanilla-type RNNs, ideas like this may be extended by training more advanced AI models, for example, based on Mamba[226] or recurrent transformers[227,228], on such tasks to test more detailed hypotheses about potential computational and architectural principles that could optimize performance. Examples that go into this direction are works that explore the commonalities of sensory representations across visual cortices and in convolutional neural networks[229] and between transformer-based language models and neural activity in cortical language areas[230], or between transformers and activity within the hippocampal formation[231]. While sometimes AI models are used only loosely as analogies for certain brain areas or functions, these links may be tightened by training on the same type of data and tasks as used for the animals, and extracting the key components from the AI models or training protocols that carry the most explanatory power. Along similar lines, computational neuroevolution offers an interesting approach for identifying key neural motifs and plasticity regimes that support adaptive behaviour, by systematically uncovering biologically plausible solutions and parameter ranges under task-relevant constraints, thus generating specific, testable hypotheses for neuroscience[206,232].

Finally, DST has not only been a major workhorse in theoretical neuroscience for a long time[233] but also is increasingly used in

ML and AI to understand the behaviour of models[104–107] and training algorithms[108–111]. The beauty of DST is that it offers a common and powerful mathematical framework for translating processes in AI systems and in brains into the same formal language, revealing functional and computational commonalities that transcend architectural and hardware differences[233]. A helpful technology in this context could be recent AI tools for dynamical systems reconstruction (DSR): DSR models learn from a set of observed time series, like multiple single-unit recordings and concurrent behavioural responses, a generative surrogate model of the underlying dynamical processes[233–237] (Fig. 5). In principle, DSR models could be any type of neural network architecture, like RNNs or autoregressive transformers, embedded within modality-specific encoder–decoder architectures[233], but for matters of mathematical tractability and interpretability they are often kept intentionally simple[235–238]. These models are then trained by specific control-theoretic procedures that help to capture long-term statistical and geometrical aspects of the dynamics[106,238,239]. The idea is that this delivers a formal surrogate model for analysing the dynamical and computational mechanisms underlying the animal's behaviour and physiology in depth. However, one may push this idea one step further and, using these techniques, train any modern AI architecture on neurophysiological activity while animals perform specific cognitive tasks, thereby directly inheriting the underlying neuro-dynamical and computational principles for solving such tasks from the data. This would move beyond copying just abstracted principles of brain function, as typical for most neuro-inspired approaches so far (Table 1). Moreover, ML methods for inferring synaptic plasticity rules directly from neurophysiological and behavioural observations have been advanced recently[240–242], without the detour of explicit computational model construction and subsequent experimental testing. These methodologies may thus not only help to tighten the link between ML and AI and neuroscience research, but could potentially provide means for direct, automated translation of dynamical and plasticity mechanisms governing task performance in the brain into AI models and training algorithms.

In conclusion, we advocate, first, to align task designs in neuroscience and in AI more closely; second, to train more advanced AI models on neuroscience tasks in search of functional insights; and, third, to use DSR models to transfer dynamical and plasticity mechanisms from data directly into AI models.

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Acknowledgements

D.D. appreciates funding from the German Science Foundation (DFG) via grants Du 354/14-1 (within the research cluster FOR-5159 dedicated to prefrontal flexibility), Du 354/15-1 and Du 354/18-1. B.A. was supported by the Intramural Research Program of the NIMH (ZIA MH002928-01). G.K. acknowledges funding from the Hector II foundation. We thank M. Kaden for her help with the figures and references.

Competing interests

The authors declare no competing interests.

Additional information

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Peer review information *Nature Machine Intelligence* thanks Andrea Soltoggio, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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